RELAY

Incident triage for logistics

BUSINESS SUMMARY / DIAGNOSTIC RETEST / 22 OCTOBER 2026

The decision

Keep routing **human-confirmed**. Text / Sensitive is the local demo default. The revision corrects the observed cold-chain failure, but this retest reuses previously inspected cases. New independent validation is needed before operational use.

Problem and working prototype

Relay proposes teams and urgency for logistics incidents, flags uncertainty and records coordinator decisions. The user-observed RLY-82374 failure reported 19 C against a 2-8 C limit. Shared evidence checks now compare stated limits and readings, handle cooling failure and distinguish controlled records or verified sensor corrections. Active exposures receive urgent human review. No real dispatch is sent.

Evidence

Two approaches: rules and trained TF-IDF centroids, with shared cold-chain protection. There are 300 synthetic development reports (240 training, 48 training families) and 60 separate reference scenarios. A fresh-context agent judged the latter using only text and guide. Labels and the original first test remain unchanged. The current rerun is diagnostic, not new unseen evidence.

Approach / policy	Macro F1	Urgent recall	Review	Cost*	p95 ms
Rules / Standard	0.601	31%	32/60	1188	0.133
Rules / Sensitive	0.601	94%	56/60	198	0.127
Text / Standard	0.683	81%	24/60	398	0.052
Text / Sensitive	0.683	100%	45/60	69	0.045

*Relative cost with miss penalty 100 for every row. Category and urgency each have 56 known labels; urgent recall uses 16 urgent cases. Four unknowns in each field remain unknown. Review uses all 60 reports. p95 excludes HTTP, startup and human decisions.

Changed requirement and workload

At the changed missed-urgent penalty of 100, Text / Standard to Sensitive changes misses from 3 to 0, reviews from 24 to 45, and cost from 398 to 69. Text / Sensitive still has 6 false urgent alerts, 2 wrong eligible routes and 1 bypassed required review. Rules / Sensitive misses 1 urgent case. These are remaining failures.

Economics, limits and next step

Both classifiers incur **USD 0 in inference API calls**; compute, development/review services and labor are unmeasured and excluded. Relative assumptions: review 1, wrong eligible route or bypassed required review 6 once per report, false urgency 2, missed priority 30 or 100 even during review. These are scenario assumptions, not savings. Reused cases, synthetic data, shared tooling and guide ambiguity limit the evidence.

Next: obtain representative operator reports and domain review, test the next candidate on a new set, and measure review capacity and escalation outcomes. No field trial has occurred.

Evidence: evidence/iterations/temperature-v3/retest.json; EVALUATION.md. Manual prototype testing and feedback: dansyar (author-reported). Software, automated checks and evaluation: OpenAI Codex. Reference labels: separate Codex agent. Full roles in ATTRIBUTION.md.